

Color-Based Image Salient Region Segmentation Using Novel Region Merging Strategy

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Abstract—In this paper, we propose a novel unsupervised algorithm for the segmentation of salient regions in color images. There are three phases in this algorithm. In the first phase, we use non-parametric density estimation to extract candidates of dominant colors in an image, which are then used for the quantization of the image. The label map of the quantized image forms initial regions of segmentation. In the second phase, we define salient region with two properties; i.e., conspicuous; compact and complete. According to the definition, two new parameters are proposed. One is called “Importance index”, which is used to measure the importance of a region, and the other is called “Merging likelihood”, which is utilized to measure the suitability of region merging. Initial regions are merged based on the two new parameters. In the third phase, a similarity check is performed to further merge the surviving regions. Experimental results show that the proposed method achieves excellent segmentation performance for most of our test images. In addition, the computation is very efficient.

Index Terms—Dominant color, importance index, merging likelihood, nonparametric density estimation, salient region.

I. INTRODUCTION

IMAGE segmentation partitions an image into nonoverlapping regions, which ideally should be meaningful for a certain purpose. Thus, image segmentation plays an important role in many multimedia applications. For example, on the internet and digital museum, huge collections of images and videos need to be catalogued, ordered, and stored in order to efficiently browse and retrieve visual information. Generally, color and texture are the most important and widely used low-level attributes for content-based visual information retrieval. Therefore, the use of low-level visual features to retrieve relevant information from image and video databases has received much attention in recent years. For last two decades, many content-based image retrieval systems have been established [1], [2]. According to the observation from extensive experiments, we found that low-level features based natural images matching using overall image similarities is often too crude such that too many unrelated images are retrieved, and consequently their performances are unsatisfactory. Therefore, it is required to segment an image into some form of perceptually relevant regions that allow the users to

identify the salient and semantically meaningful image regions, and then region-based matching can be effectively achieved. Nevertheless, the semantic segmentation of an image is still a challenging problem.

A trade off solution to narrow down the gap between low-level features and high-level human perception is to use spatial local features instead of global features of images. This means that an image should be partitioned into perceptually relevant regions, which might not correspond to objects, and then the features for matching are extracted from the segmented regions instead of the entire image. Therefore, developing a suitable image segmentation technique, which effectively partitions image into salient regions, is an important issue.

Besides the semantic gap, the illumination variation is also an annoying problem on image segmentation. Several methods have been proposed to deal with this problem [3], [4]. In our work, we proposed a simple solution in Section III-B2 to address this problem.

Because there is no formal definition for image segmentation, it is very difficult to propose a semantic index to measure a given segmentation quality. Therefore, the goal of image segmentation is very application oriented. Automatic segmentation in still image has been investigated for many years. The existing segmentation techniques can be mainly divided into the following approaches.

- 1) Histogram-based methods [5], [6]: generally deal with gray-level images, which is with 1-D histogram. However, color images are usually represented by 3-D histogram no matter what color spaces are selected. Hence, to select a global threshold or dominant color in 3-D space is a difficult task.
- 2) Region-based methods [7]–[10]: collect similar pixels into a same region according to some predefined homogeneous criteria. The region growing and split-and-merge are two popular techniques for such method. However, they strongly depend on manually tuned thresholds. For region growing, it also depends on initial seeds. These drawbacks limit the segmentation performance.
- 3) Boundary-based method [11]–[13]: these methods detect region boundaries, which are called edges in general. However, the main drawback of these methods is over-segmentation, which makes further processing necessary.
- 4) Hybrid-based method [14]–[22]: The hybrid techniques integrate the region and edge information to enhance the segmentation results. However, to integrate these two features properly is a challenging problem.
- 5) Graph-based method [23], [24]: these methods typically use graph in which the nodes represent the image pixels and arcs link the neighboring pixels. The segmentation is

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achieved by minimizing the weight that cut a graph into sub-graphs. Generally, it suffers from the high computational complexity.

In [25], a method of feature-space estimation using mean shift algorithm was proposed for image segmentation. Feature space analysis is a method to find the centers of the high-density regions. The technique for representing the significant image features is based on the mean shift algorithm, which is a simple nonparametric procedure for estimating density gradients. The method can achieve over-segmentation or under-segmentation according to different parameter setting. However, the technique is very complex and some undesired results occur frequently. Recently, Pauwels *et al.* [26] proposed a “nonparametric clustering” algorithm, which maps an image from its original feature space (intensity, color, texture) to nonparametric density space. Two nonparametric measures, “Isolation” and “Connectivity”, are then introduced to decide how to merge regions. According to their experiment, the segmentation results are good. Nevertheless, their approach is computationally expensive. Deng *et al.* [27] proposed a JSEG algorithm for color image segmentation. In the first stage, colors in the image are quantized to several representative classes. Then, the image pixel values are replaced by their corresponding color class labels, thus forming a class-map of the image. In the second stage, a J value calculation is conducted on each data point of the class-map, which maps the class-map into J feature space, thus forming a J-image. A region-growing algorithm is then applied to obtain the initial regions and followed by an agglomerative method to merge the initial regions. As shown in their experimental results, they also reveal good segmentations. However, the JSEG algorithm suffers from the same difficulty, the high computational cost, as the “nonparametric clustering.”

To improve the segmentation results, many researchers analyze various properties of images by utilizing complicated formulas. This sacrifices the execution time and makes them hard to implement. The bottleneck of time consuming of those complicated algorithms limits their scope of application. Because of perception subjectivity, a good all-purpose algorithm for image segmentation does not exist. The main purpose of this paper is not to precisely segment every single object in an image but to find the salient regions that are relatively meaningful to human perception. In addition, the method that extracts salient regions should be very time efficient so that it can be applied to online applications such as region based image/video retrieval. In the past few years, some methods have been proposed for finding salient regions in images [26], [28], but the computational complexity and the segmentation results are not satisfactory. In this paper, we will propose an effective method to address these drawbacks.

In the new approach, we develop a fast dominant color-extraction scheme based on nonparametric density estimation. The new scheme automatically determines the number of dominant colors and effectively reduces the representative colors in an image. We also give a definition of region salience and accordingly develop a novel merge strategy, which takes into account not only the homogeneity but also the geometric properties of regions. Experimental results show that the proposed method has very promising performance.

We organize this paper as follows. In Section II, we briefly describe the image segmentation problems. The details of the

proposed method are given in Section III. In Section IV, we show and discuss the experimental results. Brief conclusions are given in Section V.

II. PROBLEM FORMULATION

Image segmentation partitions an image into nonoverlapping regions with respect to some specific homogeneous properties. There are several ways to define homogeneity of a region, such as color, texture or gray level, according to particular image type. Each pixel in an image will be assigned to one and only one region; therefore, region overlapping is not permitted. Usually the homogeneity among neighboring regions will be further checked, and then the homogeneous regions will be merged into one region. Because the segmentation is not well defined, some intrinsic problems exist. We briefly summarize as follows.

- 1) There is no universal homogeneity suitable for all image types. Therefore, it is hard to select a representative image feature for the measure of homogeneity.
- 2) For unsupervised segmentation, the final number of segmented regions is unknown *a priori*. There are no appropriate criteria to determine the correct number of regions in an image.
- 3) Many manually tuned parameters or thresholds need to be carefully preset, because the different setting will dramatically change the segmentation results.

In this paper, we will propose an approach of salient region segmentation to address these problems, and improve the segmentation quality. In our work, we take aim at the segmentation of natural color image; therefore, we select color as image feature because color is the most important attribute in natural images.

For salient image segmentation, the salience is a macro property of an image. In other words, a salient region can be easily identified when we see an image. As shown in Fig. 8.1(a), we can easily see a dog sitting on the grass. Therefore, the dog and the grass are salient regions, although the dog and the grass themselves are not homogeneous in color or texture. Unlike the object segmentation, salient region segmentation is not necessary to extract each object in an image accurately but viewing the whole objects as a salient region, as shown in Fig. 8.4 in which the herd of elephants is a salient region. As not requiring a priori domain knowledge about object, salient region segmentation is more feasible for applications such as region based image/video retrieval than is the object segmentation.

In this paper, we propose a new region-merging strategy based on the image salience and a new merging rule. We first calculate the “Importance index” of each region and then merge those regions with lower value of “Importance index” into one of its neighboring region according to the new merging rule. As a result, the final segmented regions satisfy the image salience. The proposed algorithm is illustrated in Fig. 1, which is divided into three phases. In the following sections, we will describe each phase of the new algorithm in detail.

III. DOMINANT COLOR EXTRACTION, IMAGE QUANTIZATION, AND REGION MERGING

For a true color digital image, there are a huge number of colors in color space no matter what the chosen color space

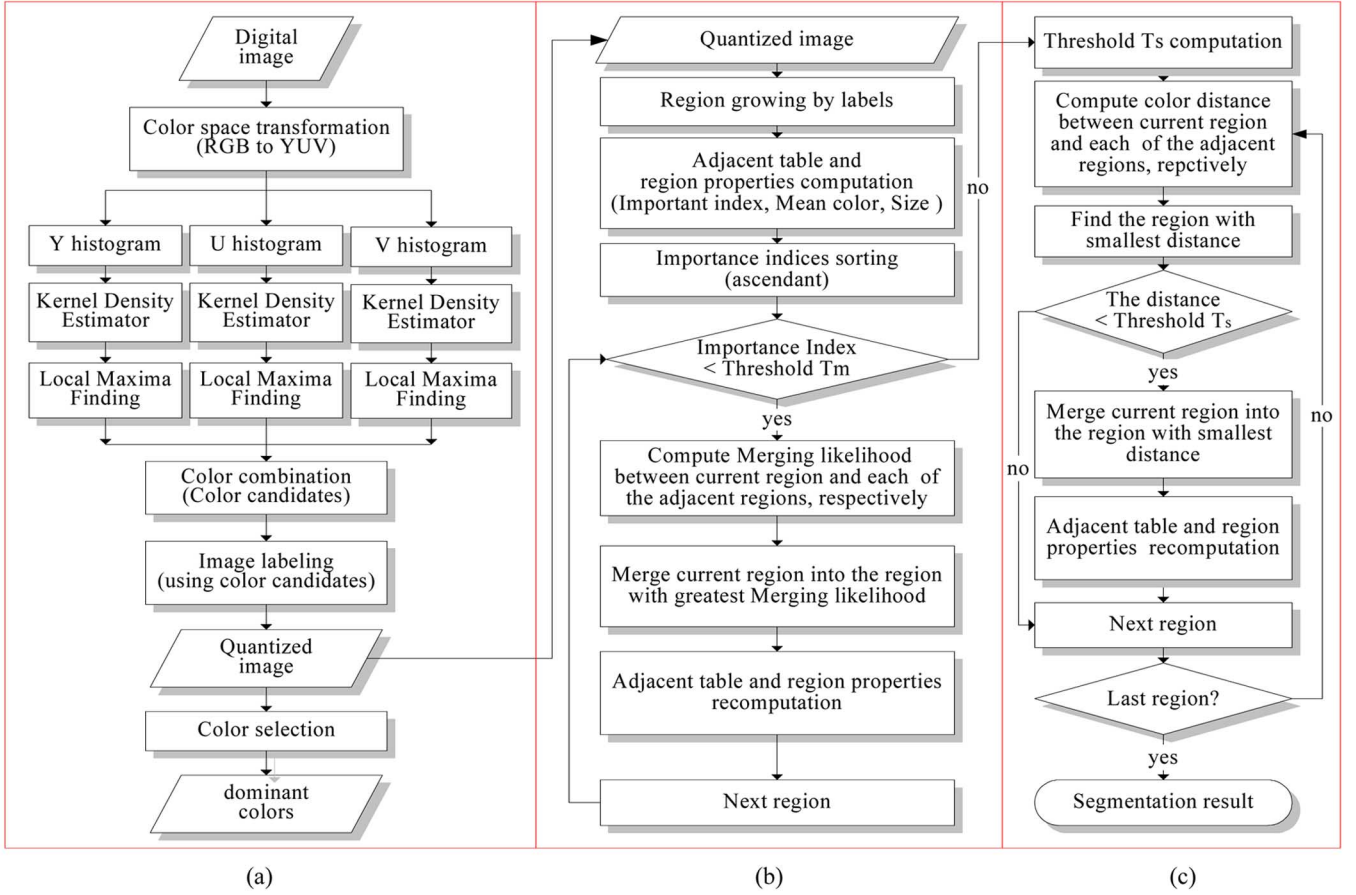


Fig. 1. Flowchart of the proposed segmentation algorithm. (a) Dominant color extraction. (b) Region merging based on merging likelihood. (c) Region merging based on color similarity.

is. The representative colors (or dominant colors) of an image, which preserve all human-relevant discriminative information, are critical for natural image segmentation. Therefore, an efficient and effective scheme to extract the representative colors is necessary.

The generalized Lloyd algorithm (GLA) [29] is the most extensively used algorithm to extract dominant colors from an image; however, its computational cost is expensive. Moreover, it suffers from three intrinsic problems.

- 1) It may give quite different kinds of clusters when cluster number is changed.
- 2) A correct initialization of the cluster centroid is a crucial issue, because some clusters may be empty if their initial centers lie far from the distribution of data.
- 3) The effectiveness of the GLA depends on the definition of “distance”; therefore, the choice of a specific distance measure may change the clustering results.

Using the splitting algorithms is a possible way to overcome the difficulties [30]–[32], because these algorithms start with arbitrary random centroids. However, they often converge to local optima and usually require high computational complexity. Hence, we propose a fast method to extract dominant colors in an image, and quantize the image by using the dominant colors.

In our work, we develop a new dominant color-extraction scheme based on nonparametric density estimation [26], [33], [34]. Given an n -dimensional dataset $\{x_i \in R^n; i = 1 \cdots N\}$,

the nonparametric density $f(x)$ is obtained by convolving the dataset with a unimodal density kernel $K_\sigma(x)$

$$f(x) = \frac{1}{N} \sum_{i=1}^N K_\sigma(x - x_i) \quad (1)$$

where σ is the bandwidth for the kernel. In our work, we selected a Gaussian kernel as

$$K_\sigma(x) = \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{n}{2}} e^{-\|x\|^2/2\sigma^2}. \quad (2)$$

To estimate the density for the entire feature space, the computational complexity of (1) is $S \times N$, where S is the size of the feature space and N is the number of all data points. When both S and N are large, the computation cost is huge. Let us give a simple example:

Considering the color as the feature space, e.g., RGB color space, there are three color channels. Each color channel is normalized from zero to 255 as integer. Consequently, there are totally 256^3 possible colors in the feature space; i.e., $S = 256^3$. For an image, each pixel is a 3-D data-point and the density estimation is conducted for each pixel. For a CIF format color image, there are 352×288 pixels in an image; thus, $N = 352 \times 288$. Therefore, the computational complexity is $256^3 \times 352 \times 288$, which is prohibitively huge.

Because the size of color space is so large, not all colors in the color space will appear in an image. Actually, most images use only a small number of colors compared with the size of color space. Hence, we make a simple modification to reduce the computational complexity. Let I denote a color image, and then we reformulate (1),(2) into (3),(4), respectively, as follows:

$$f(i_{x,y}) = \frac{1}{WH} \sum_{j=1}^W \sum_{k=1}^H K_{\sigma}(i_{x,y} - i_{j,k}) \quad (3)$$

$$K_{\sigma}(i_{x,y}) = \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{n}{2}} e^{-\|i_{x,y}\|^2/2\sigma^2} \quad (4)$$

where $i_{x,y} \in I$ is the color of a pixel at position (x,y) and $i = (i_r, i_g, i_b)$ is a 3-D color vector. Moreover, the W and H are the width and height of the image, respectively. Equation (3) estimates the nonparametric density of a given color on image plane. For an image, the computational complexity of (3) is N^2 , where $N = (W \times H)$ is the size of the image. Because $S = 256^3$ is much greater than N , the computational complexity of (3) is much less than (1). Therefore, (3) is a better solution.

Although N^2 is much less than $(S \times N)$, N^2 is still a large number for computation. To address this problem, we decompose the 3-D color space into three 1-D feature spaces. We estimate the densities on three 1-D color channels instead of one 3-D color space. Therefore, we reduce the size of feature space from $256 \times 256 \times 256$ to $256 + 256 + 256$, which is about 0.0046% of the original size.

Because the luminance and chrominance are mixed together in RGB color space, we adopt YUV color space to take advantage of decorrelating the luminance and chrominance, which will be discussed later in Section III-B2. To further speed up the density estimation, the convolution of Gaussian kernel does not directly apply to the source image plane but to the histogram of each channel. Let $h(r)$ denote the histogram of an image for one of the three color channels, where $r \in \{Y, U, V\}$. We then reformulate (1) and (2) into (5) and (6), respectively, as follows:

$$f(r_k) = \frac{1}{M} \sum_{i=1}^M h(r_i) K_{\sigma}(r_k - r_i) \quad (5)$$

$$K_{\sigma}(r_k) = \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{1}{2}} e^{-\|r_k\|^2/2\sigma^2} \quad (6)$$

where r_k is the k th level of that channel and M is the total number of levels of it. Although the computational complexity of the channel is M^2 , the computational cost is very low in that M is a small number, say 256. The density of each channel (Y, U, and V) is estimated by (5), which tremendously decreases the required processing and achieves equivalent results.

After nonparametric estimation, the density distribution for each channel is obtained. Using the gradient ascent scheme, we can easily find the local maxima. We select the local maxima of each channel and combine them to form the candidates of dominant colors. The number of candidates depends on the bandwidth σ of the Gaussian kernel and will decrease when a large bandwidth is used and vice versa. Fig. 2 illustrates the dominant color-extraction scheme. The original densities of Y, U, and V channels in Fig. 2(a) have many local maxima; therefore, it will generate too many colors to represent the image. However,

due to taking into account all influences from other data points, the nonparametric densities revealed a much smoother distribution, and reduced the number of local maxima. As shown in Fig. 2(b), each of the three channels has three local maxima. Consequently, we get $3 \times 3 \times 3 = 27$ candidates of dominant colors, as shown in Fig. 2(c).

To avoid over-smoothing of the densities, a relatively smaller bandwidth is preferred. A smaller bandwidth may result in a larger number of candidates of dominant colors but will not cause any problem to image pixel assignment in our work, because we assign the image pixels to one of the nearest candidates by a fast mapping algorithm, which will be discussed in the next subsection.

A. Image Pixel Assignments

One common step of image quantization algorithms is to assign the color of each pixel of an image with the nearest color in the color quantization table. Colors are cardinal triplets and therefore no order relation between each other. Consequently, they cannot be sorted. To find the nearest (most similar) color of a given color from the color quantization table, we should compute the distance between the given color and each of the colors in the quantization table using a specified distance function, e.g., Euclidean distance, and then select the minimum. Suppose that there are Cn colors in the table, we have to do the distance computation Cn times for each pixel of the image, and then find the minimum as our target. When both the number of colors and the image size are large, the assignment becomes a burden of time. Fortunately, our dominant color-extraction scheme enables us to do the assignment using a fast algorithm. In our dominant color-extraction scheme, mentioned in the previous subsection, the candidates of dominant color are all the combinations of the local maxima of the three color channels. This characteristic allows us to build a nearest-color mapping table, which transforms the distance computation to table lookup. The nearest-color mapping table of Fig. 2 is shown in Fig. 3, in which each channel is partitioned based on the local maxima of it, and then the nearest local maximum is assigned to the nearest value of each corresponding original value in the table. Unlike the distance computation algorithm, the table lookup algorithm needs only one step to obtain the nearest color, irrespective of the number of colors in the candidate list. When a color is given, the value of each channel of the color is used as an index to obtain the nearest value on that channel. For example, a color ($y = 25, u = 220, v = 148$) is given, by looking up the nearest-color mapping table in Fig. 3, we got the nearest dominant color ($y = 67, u = 123, v = 151$). No matter what the color is given, we can always find one and only one nearest color that exists in the candidate list. After the nearest color has been obtained, we also need to know the corresponding label. Hence, the values of the three components of each candidate are concatenated forming a single scalar. The candidate list then becomes a scalar list, which can be sorted, enabling a binary search. This reduces the time complexity of pixel assignment for a single pixel from Cn to $(\log_2 Cn) + 1$ for the worst case, in which the table lookup step is counted for 1.

After the pixel assignments, each pixel in image has been replaced by the nearest candidate. Consequently, a quantized

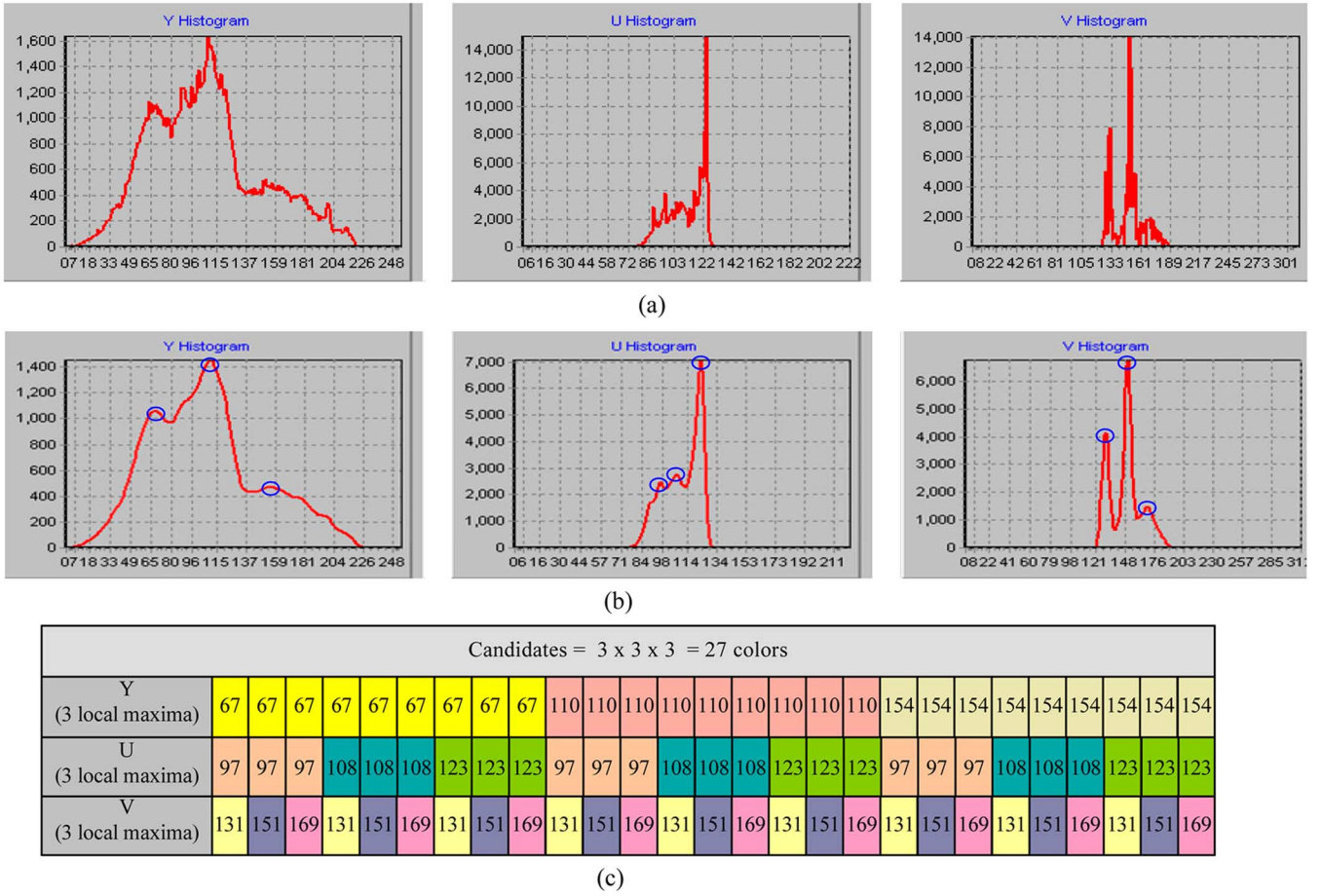


Fig. 2. Example of the dominant color extraction. (a) Original densities. (b) Nonparametric densities. (c) Color combinations.

color image is obtained and a label map is created as well. The candidates of dominant color are all possible combinations of each color channel's local maxima. Therefore, we can simply eliminate those candidates with lower pixel count, and keep the rest as dominant colors.

We apply the region-growing algorithm on the label map of the quantized image to obtain initial regions. Some of them may be very small and less important. Therefore, not all initial regions are salient. In the following, we will define the salience of image region, calculate the region importance accordingly and then develop a new merging strategy to form the salient regions.

B. Region-Merging Strategy

For salient image segmentation, the salience is a macroscopic property of an image. In other words, the salient region can be easily identified when we see an image. In our definition, they should have two properties as follows.

- *Salient regions should be conspicuous.* In general, when we look at an image, the larger regions may contain more complete information for human's perception than that of small regions; therefore, they will capture our attention with most probability. For instance, looking at Fig. 8.1(a) we may readily find a dog, and the grass as its background. However, we may not pay much attention to the eyes of the dog, or how many flowers there are in the background, because they are not with complete information. For Fig. 8.3(a), we

may perceive that there exist an elephant, the grass, and the sky, but not the tusks. Thus, we can conclude that a salient region with conspicuous should be big enough.

- *Salient regions should be compact and complete.* When an image is processed by quantization and region growing, the obtained regions are dissimilar to each of their adjacent regions from the viewpoint of homogeneity, regardless of the sizes of them. A large region may contain many small regions or holes inside it. Holes inside a region violate the region completeness and should be combined to form a more complete and meaningful region. On the other hand, there may have some small regions with same color label but distributed unconnected in the entire image, shown as Fig. 4. Although, the total size of these regions is big enough, they are not compact. Thus, they are not salient regions.

The region-merging strategy plays a critical role for image segmentation. In the literature, most merging rules are mainly based on the criteria of homogeneity. Generally, after the region growing, the initial regions themselves are homogeneous, but they are dissimilar to each of their adjacent regions. Thus, the homogeneity driven approaches cannot achieve the requirement of salient region. We need new rules to check each region and merge the unsatisfactory ones.

In our work, two new parameters are proposed. One is called "Importance index", which is used to measure the importance of a region, and the other is called "Merging likelihood", which

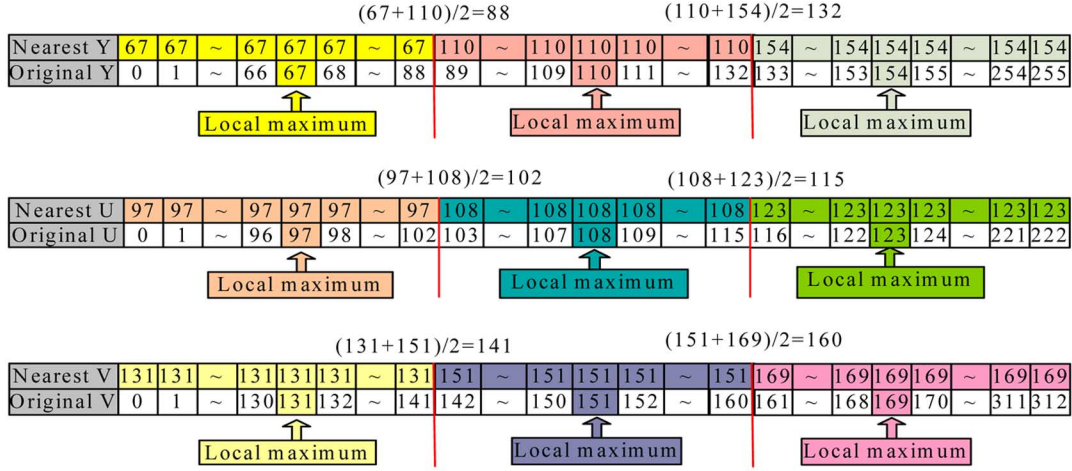


Fig. 3. Nearest color mapping table of Fig. 2.

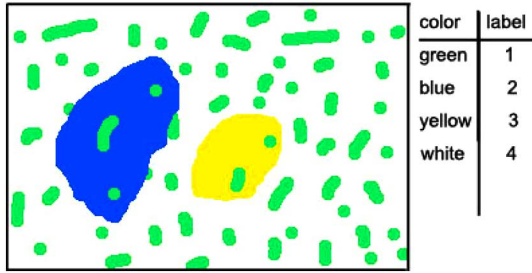


Fig. 4. Example of noncompact regions.

is utilized to measure the suitability of region merging. Whether a region should be merged mainly depends on its “Importance index”, and where it should be merged into depends on the “Merging likelihood” between the region and each of its adjacent regions. In the following, we will define the “Importance index” and the “Merging likelihood”.

Importance Index Computation: “Importance index” is an indicator of importance for every single region in an image. A region with a higher value of importance index indicates that it is more important than those with a lower value.

Based on our definition of salience, the importance index should reflect the conspicuousness, compactness and completeness of a region. Intuitively, the first property can be easily examined by the ratio of the region size to the image size such that a bigger region will have a higher ratio. However, it is not easy to express the second characteristic by a simple mathematical expression, because it is actually a high-level perceptual description. We cannot know if there are holes in a region before the region has been detected. Even though a region has been detected, it takes a lot more work to find holes in it. Nevertheless, by changing the point of view from high-level perception to low-level features, we may get an easier and acceptable solution. When an image is quantized by the dominant colors, many regions may have a color same with others. A color appearing only in a single region should be more important than those colors appearing in multiple regions since it is unique, and the corresponding region should have a higher value of importance. Therefore, the second characteristic can be expressed as

the ratio of the region size to the largest size of all regions that have the same color label as the region. Obviously, a unique region will have a higher value.

Finally, we define the importance index as the multiplication of these two ratios. The mathematical formulation of “Importance index” is expressed as

$$Imp(R_j^i) = \left(\frac{N_{R_j^i}}{\sum_{j=1}^m N_{R_j}} \right) \times \left(\frac{N_{R_j^i}}{Max(N_{R_i})} \right) \quad (7)$$

where

- R_j^i region with color label i , region index j ;
- $Imp(R_j^i)$ importance index of R_j^i ;
- $N_{R_j^i}$ number of pixels of R_j^i ;
- $Max(N_{R_i})$ number of pixels of the largest region with color label i ;
- $\sum_{j=1}^m N_{R_i}$ total number of pixels of the image (image size);
- m number of the initial regions.

If the importance index of a region is less than the merging threshold Tm , then the region is called a less important region; otherwise, an important region. Less important regions must be merged into one of their adjacent regions while the important regions are not necessary to be merged but a further check is proceeded to decide if they should be merged. The Tm is a very important parameter, which controls the degree of merging, but is not sensitive because the merging is processed by two phases. Based on our extensive observation, the Tm is set proportional to the importance index of the second important region. We found, for most images, the largest region in an image is the background, and the second large region is usually the main object with significant size. In general, the second large region corresponds to the second important region. Therefore, the importance index of second important region is more suitable to be the merging threshold than the largest one.

Merging Likelihood Computation: We take into account both homogeneity and geometric properties of regions to compute

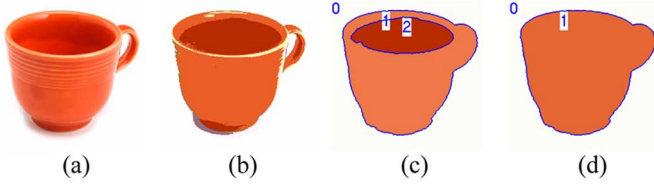


Fig. 5. Example of segmentation results affected by variation of illumination. (a) Source image. (b) Quantized image. (c) Segmentation result using unweighted Euclidean distance function. (d) Segmentation result using weighted Euclidean distance function.

the “Merging likelihood” between regions. Three factors are considered to express the mathematical form of the merging likelihood.

- *Color distance between regions.* The first factor is the color distance between regions. Color distance is the most common factor used in similarity measurement and usually the Euclidean distance is utilized as the distance function. The proposed distance computation is different from the general Euclidean distance. In our method, we adopt a weighted Euclidean distance to reduce the influence of illumination. Since the human eye is more sensitive to luminance than chrominance, humans can easily tell the differences affected by the variation of illumination on an object. Nevertheless, for an image segmentation task, the variation of illumination may produce wrong segments. As shown in Fig. 5(a), there is a cup in the image. Although we can easily identify some portions of the cup are darker and some are brighter, we know that there is only a cup and nothing else. In RGB color space, the color of darker area and the color of brighter area are different in each of the three color channels. However, in YUV color space, their difference is mainly on Y channel (luminance) and the U and V channels (chrominance) remain the same. Therefore, we conclude that the difference between darker area and brighter area is caused by the variation of illumination. Consequently, by reducing the influence from Y channel (luminance) we can improve the segmentation results. Hence, we adopt a weighted Euclidean distance with a lower weighting for Y and a higher weighting for U, V channels. In [35], an extensive analysis of justification for the weightings is proposed. In our work, we define the weighted Euclidean color distance between two regions as shown in (8), at the bottom of the page, where the w_Y , w_U and w_V are the weightings for Y, U and V channels, respectively. Fig. 5(c) and 5(d) demonstrate the segmentation results of our segmentation algorithm using unweighted and weighted Euclidean distance, respectively. Obviously, Fig. 5(d) is more consistent with human perception.

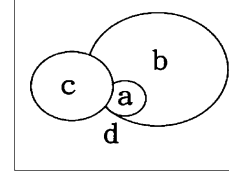


Fig. 6. Example of region merging concerning boundary length between regions.

- *Boundary length between regions.* The second factor is the boundary length between regions. We use an example to explain the principle. Fig. 6 shows an image with four regions. Suppose that region “a” should be merged into one of the other three regions. Instinctively, region “a” is more likely to be part of region “b” because region “a” is on the inside of region “b”. It is quite easy for human to figure out this situation but not easy for computational algorithm. However, it is easy to count the boundary pixels between region “a” and each of other regions for computer. Because the longer the boundary length between two regions the stronger the connection between them, the region that has the longest boundary length between itself and region “a” should have the highest priority to be the merging region. In this case, the boundary length between region “a” and region “b” is the longest one, and therefore region “a” should be merged into region “b”, which is consistent with our intuition.
- *Region sizes of neighboring regions.* The third factor is the region sizes of neighboring regions. During the merging stage, a region with smallest importance index will be merged into one of its neighboring regions. Should a region be merged into the largest neighboring region or the smallest neighboring region? According to our extensive observation, for most images, the largest region in the image is the background. If we merge a region into its largest neighboring region, there is a higher probability that we will merge the region into the background. If we merge a region into its smallest neighboring region, the merging region may still be small and will have a lot more chances to be merged into a correct region. Therefore, the smallest neighboring region should have the highest priority to be the merging region, concerning the region sizes only.

To define the merging likelihood between regions, the following assumptions are given.

- 1) Assume a is a region to be merged and b, c, d are its neighboring regions, $S = \{b, c, d\}$
- 2) Let $CD(a, k)$ denote the color distance between region a and region k , $k \in S$

$$CD(R_1, R_2) = \sqrt{w_Y (y_{R_1} - y_{R_2})^2 + w_U (u_{R_1} - u_{R_2})^2 + w_V (v_{R_1} - v_{R_2})^2} \quad (8)$$

3) Let $BL(a, k)$ denote the boundary length between region a and region $k, k \in S$

4) Let $RS(k)$ denote the region size of region $k, k \in S$

Based on the assumption above, “Merging likelihood” is defined as

$$ML(a, k) = w_1 \left(1 - \frac{CD(a, k)}{\max_{k \in S} (CD(a, k))} \right) + w_2 \left(\frac{BL(a, k)}{\max_{k \in S} (BL(a, k))} \right) + w_3 \left(1 - \frac{RS(a, k)}{\max_{k \in S} (RS(k))} \right) \quad (9)$$

where w_1 , w_2 and w_3 are the weights for color distance, boundary length and region size, respectively.

In merging likelihood, we take into account both color homogeneity and geometric properties between regions. After the color quantization and initial region extraction, the regions are generally not so similar in color. Therefore, the geometric properties, i.e., the region size and the boundary length, should dominate in merging likelihood. Therefore, the parameters should satisfy the condition $w_1 < w_2 + w_3$.

From the definition above, we can easily find that a region with a smaller color distance, longer boundary length, and smaller region size will produce a higher value of merging likelihood. Finally, region a should be merged into the region with largest merging likelihood.

Similarity of Important Regions: As mentioned in Section III-B.1, the less important regions will be merged into one of their adjacent regions according to the merging likelihood. After the merging process is complete, all the surviving regions are important. However, they should be further checked because some important regions, which have no connection between each other before merging, may become adjacent due to the less important regions located between them have been merged. Therefore, important regions may still be similar to their adjacent regions and should be further merged. The further checking criterion is to measure the color similarity with an adaptive threshold T_S . We define the adaptive threshold T_S as in (10)

$$T_S = \text{Mean}_{\text{dis}} \times W_S, \\ W_S = \begin{cases} 0.7, & \text{if } W_{\text{dynamic}} > 0.7 \\ 0.55, & \text{if } W_{\text{dynamic}} < 0.55 \\ W_{\text{dynamic}}, & \text{otherwise} \end{cases} \\ W_{\text{dynamic}} = 1 - e^{-\left[\frac{\text{Var}_{\text{dis}}}{C_1}\right]} \quad (10)$$

where

Mean_{dis}	average of color distances between each of the surviving regions;
Var_{dis}	variance of the color distances;
C_1	constant, which controls the output range of the exponential function.

If the color distance between two connected important regions is less than T_S , they are similar and should be merged. Otherwise, they should be separate. According to our extensive

experiments, the threshold is between about 0.55 to 0.7 of mean distance (i.e., Mean_{dis}). Therefore, we set 0.55 and 0.7 as the lower and upper bound of W_S , respectively. We use C_1 to control the dynamic range of the weight value between 0.55 and 0.7. In our work, we set C_1 as 110.

Physically, when those important regions are similar in color, we should use a relatively lower threshold for the merging process to prevent over-merging. On the contrary, when they are dissimilar, we should use a relatively higher threshold such that those relatively similar regions will be merged. We can learn from (10) when the mean and variance of the color distances are small, which means they are similar, we will have a smaller T_S and vice versa. Therefore, (10) satisfies the merging requirement.

The Process of Segmentation: In order to make it easier for readers to understand the proposed algorithm, we summarize it by an example with some intermediate results, shown in Fig. 7, as follows.

- 1) Given a target image, shown in Fig. 7(a).
- 2) Convert the color space of the image from RGB to YUV.
- 3) Create histograms of the image for each channel, shown in Fig. 2(a).
- 4) Estimate the nonparametric densities of each channel using (5) and (6), shown in Fig. 2(b).
- 5) Obtain the candidates of dominant colors by combining local maxima of the three channels, shown in Fig. 2(c).
- 6) Quantize the image, shown in Fig. 7(b), using the candidates of dominant colors by looking up the nearest-color mapping table in Fig. 3, and create a label map.
- 7) Apply region-growing algorithm on the label map to obtain initial regions, shown in Fig. 7(c).
- 8) Compute importance index and region properties for each initial region and sort the regions by importance index with ascendant order.
- 9) Merge all less important regions in order based on the rules of “Merging likelihood”; the surviving regions are shown in Fig. 7(d).
- 10) Compute color distances between each of the surviving regions and compute the adaptive threshold T_S , shown in Table I.
- 11) Merge the connected regions whose color distance is less than T_S , and the result is shown in Fig. 7(e) and (f).

When a region is merged into another region, three things need to be done: 1) change the region index of the merged region to the merging region; 2) recompute the region properties (mean color, region size and boundary length) of the merging region; and 3) modify the adjacent table of regions. The mean color can be computed by a recursive form of weighted average, the new region size can be obtained simply by a summation, and the boundary length and adjacent relation between the merged region and its connected regions will propagate to the merging region. No pixel scan is needed. Therefore, the merging process is very computational efficient.

Besides, there is no actual region index changing on the region map during the iteration stage of region merging. Only a record is made to track the merging. Table II gives a simple and arbitrary example of the record to illustrate the mechanism. In the beginning, there are ten initial regions. After the iteration

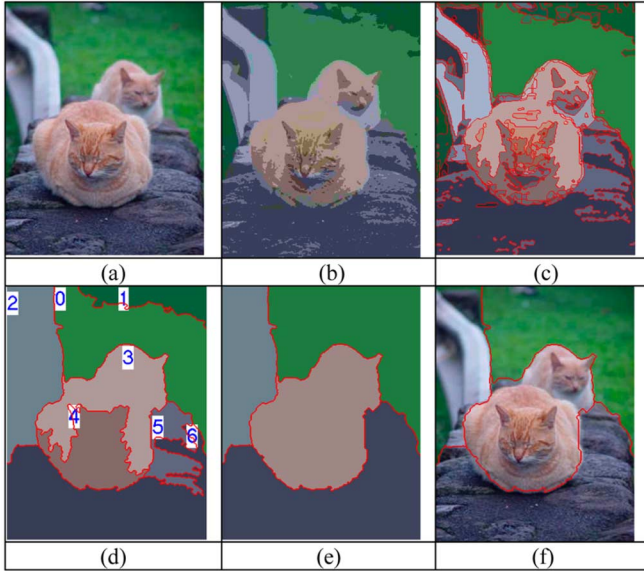


Fig. 7. Intermediate results of the proposed algorithm. (a) Source image (b) Quantized image. (c) Initial regions. (d) Surviving regions represented in mean colors with region number. (e) Result (after further-merging) represented in mean colors. (f) Final Segmentation result.

stage of merging, three regions survived and a mapping table is created. Then, by which the initial regions can be mapped to the corresponding final regions. Finally, the region map can be modified on a single pass, and then we get the segmentation results. This eliminates the redundancy of region index changing during iteration stage, and thus speeds up the process.

IV. EXPERIMENTS AND DISCUSSIONS

We selected two image databases to perform the experiments. The Corel Image Database is widely used in the simulation of image retrieval. Therefore, we chose 100 from the database as our test set. Generally, there is a clear subject in an image. Thus, most selected images have conspicuous regions and satisfy the definition of salience.

Furthermore, Malik's group at Berkeley has done an excellent work on segmentation [36]–[39]. We also download their database for testing our algorithm. They divided the images in this database into a training set of 200 images, and a test set of 100 images. The contents of this database are in a wide range. From a simple foreground with simple background to complex foreground with complex background, it is a much more exacting test for any algorithm.

We use the 200 training images to tune our parameters and the identical setting is used for entire simulations.

A. Parameter Setting and Tuning

Some parameters in our algorithm need to be preset.

- 1) The bandwidth σ of the convolution kernel: $\sigma = k \times SD$, where SD denotes the standard deviation of the channel, and $k = 0.15$ for Y and 0.3 for U, V channels. For each channel, if only one local maximum is obtained then a

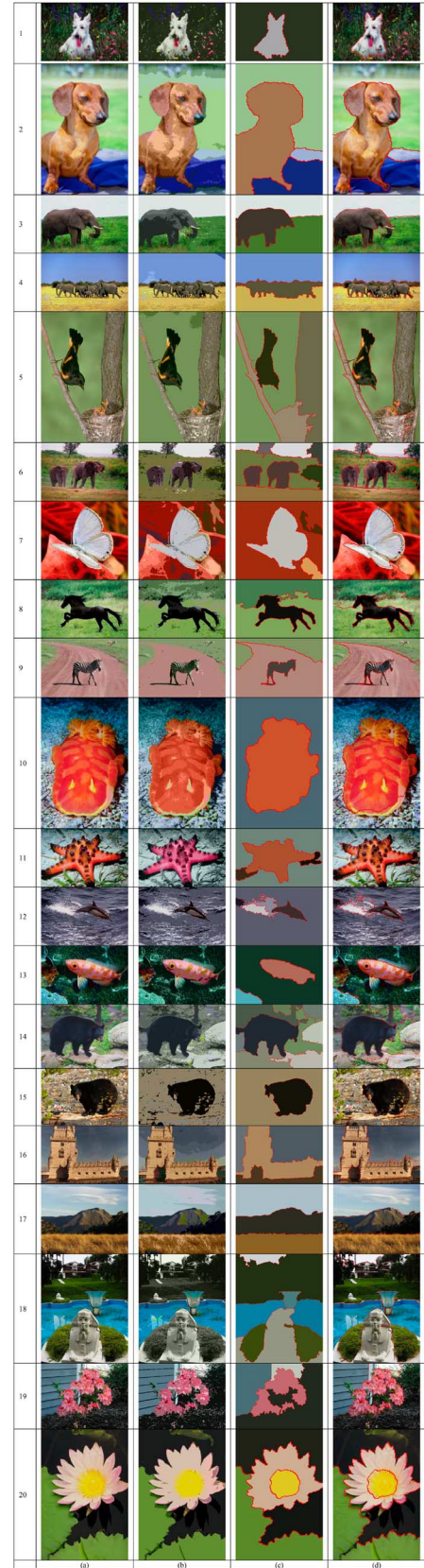


Fig. 8. Experimental results of our database. (a) Source images. (b) Quantized images. (c) Segmentation results represented in mean colors. (d) Segmentation results.

slightly smaller value of k will be applied to re-estimate the distribution automatically to prevent over-smoothing.

TABLE I
COLOR DISTANCE BETWEEN EACH OF THE SURVIVING REGIONS OF FIG. 7(D)

	R1	R5	R4	R2	R3	R0	R6
R1		*****	*****	*****	*****	2.35E+01	*****
R5	*****		3.72E+01	*****	4.25E+01	6.31E+01	2.04E+01
R4	*****	3.72E+01		4.40E+01	2.33E+01	*****	4.41E+01
R2	*****	*****	4.40E+01		4.26E+01	5.36E+01	3.32E+01
R3	*****	4.25E+01	2.33E+01	4.26E+01		7.47E+01	*****
R0	2.35E+01	6.31E+01	*****	5.36E+01	7.47E+01		6.69E+01
R6	*****	2.04E+01	4.41E+01	3.32E+01	*****	6.69E+01	

Note: (1) $T_S = 2.81E+01$ (2) ***** indicates no connection
(3) The cells with blue mark indicate that the value is less than T_S

TABLE II
EXAMPLE RECORD OF REGION MERGING

Initial regions	0	1	2	3	4	5	6	7	8	9
Be merged into	-1	3	3	5	-1	4	-1	6	6	7
Final regions	0	4	4	4	4	4	6	6	6	6

Note: -1 denotes no merge

- 2) The merge threshold T_m : $T_m = 0.2 \times \text{Imp2}$, where the Imp2 denotes the importance index of the second important region.
- 3) The weightings w_Y, w_U, w_V for color distance computation: $w_Y : w_U : w_V = 1 : 10 : 10$, and normalized, i.e.,

$$w_y = \frac{w_Y}{w_Y + w_U + w_V}, w_u = \frac{w_U}{w_Y + w_U + w_V}$$

$$w_v = \frac{w_V}{w_Y + w_U + w_V}, w_y + w_u + w_v = 1.$$

- 4) The weightings w_1, w_2, w_3 for merging likelihood computation: $w_1 = 2, w_2 = 1.75$, and $w_3 = 1.5$.
- 5) The constant C_1 for threshold T_S computation: $C_1 = 110$, as mentioned in Section III-B.

We obtain these parameters by using a systematic approach, which is briefly described as follows. Based on our experiences, we select an initial set of parameters at the beginning, and then execute the following steps.

- 1) The 200 training images are segmented using the initial set of parameters.
- 2) Some of the unsatisfactory ones are selected for further analysis.
- 3) We divide the parameters into three groups as image quantization, region merging and further merging. For each selected image, we tune the parameters of each group separately. The three tuning stages are as follows.
 - 1) For image quantization stage, the parameter k , which is used to compute the bandwidth σ of the convolution kernel for each color channel, needs to be adjusted. Generally, the resulting candidates of dominant colors are not more than 300; the value of k is suitable. We record the range of k .
 - 2) For region-merging stage, three parameters w_1, w_2, w_3 need to be adjusted. Observing the merging result, we tune the parameters until most regions are complete without containing other regions; the values of w_1, w_2, w_3 are suitable. We also record the parameters' value.
 - 3) For further merging stage, the parameter C_1 needs to be adjusted. Observing the final segmentation, if over-merging happens, we increase C_1 . If under-merging

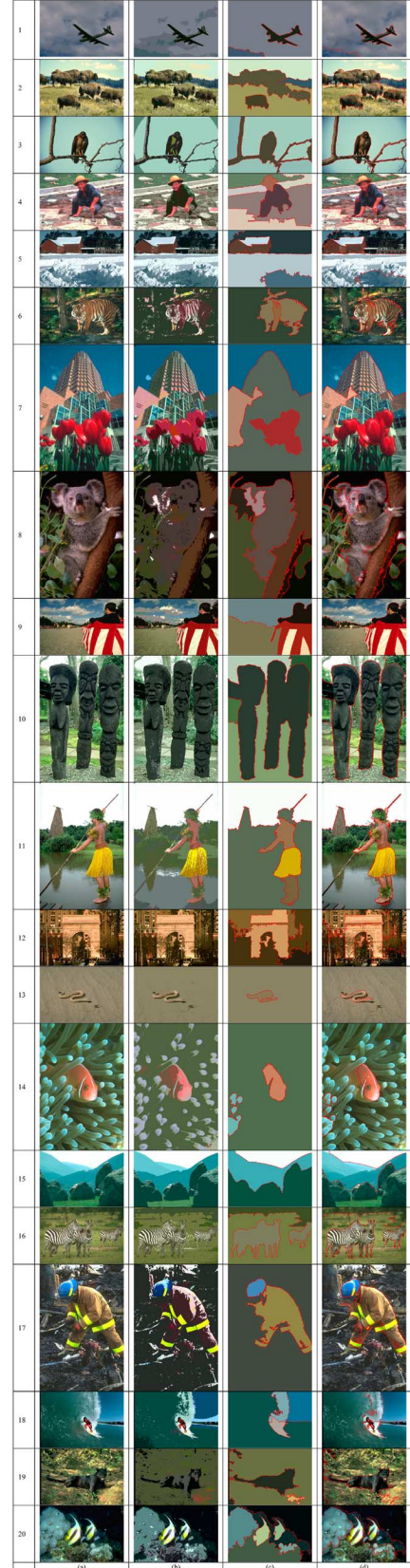


Fig. 9. Experimental results of Malik's database. (a) Source images. (b) Quantized images. (c) Segmentation results represented in mean colors. (d) Segmentation results.

happens, we decrease C_1 . We can easily find the tolerance of C_1 for the image. We record the range of C_1 .

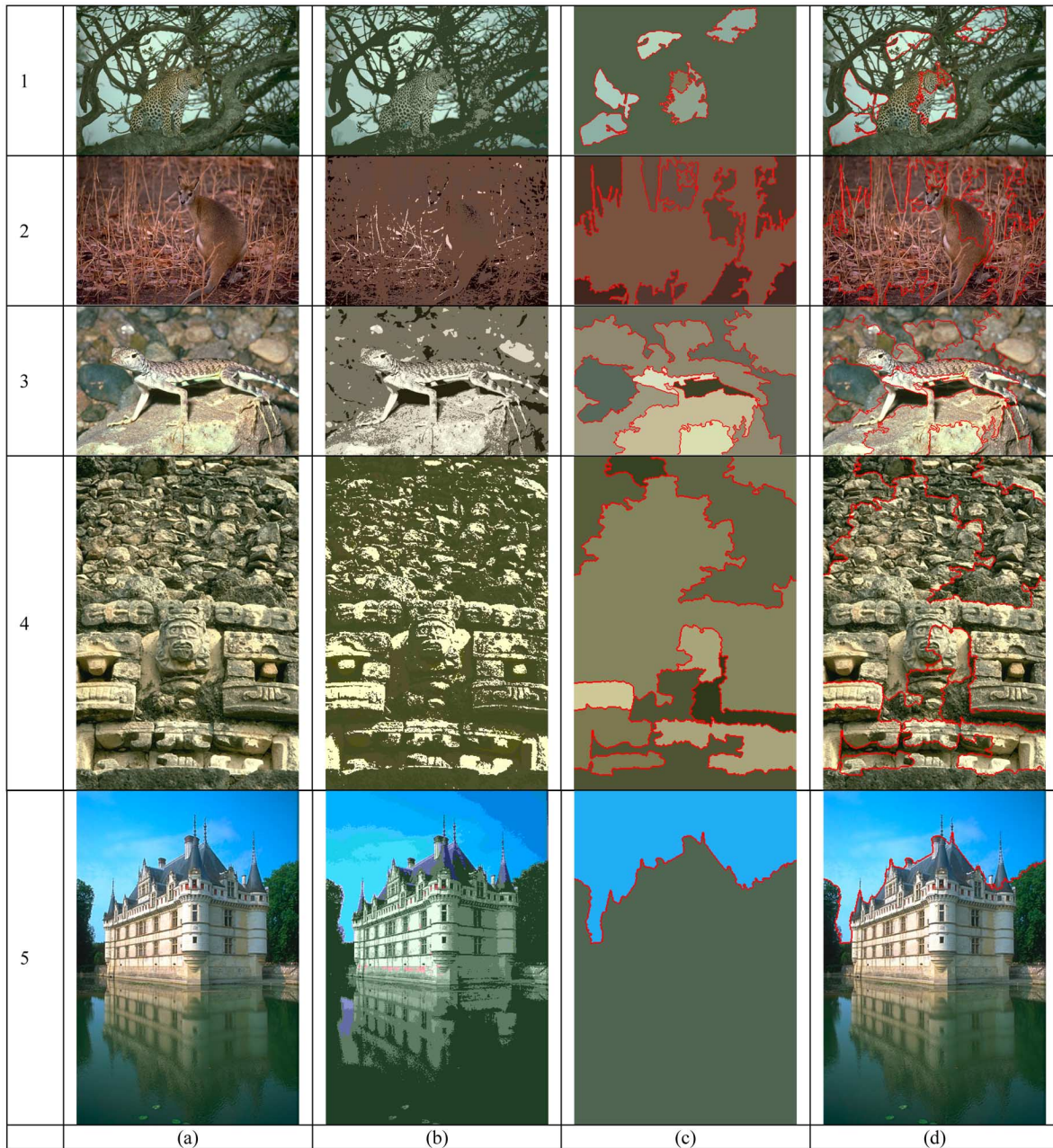


Fig. 10. Typical failure cases. (a) Source images. (b) Quantized images. (c) Segmentation results represented in mean colors. (d) Segmentation results.

- 4) Select a new set of parameters within the ranges.
- 5) Using the new parameters, we segment the 200 training images again.
- 6) If the segmentation results are satisfactory or cannot be better, stop the tuning process; otherwise, go to step 2.

For each stage, a slight modification of the parameters' values will not dramatically change the segmentation results. Because each stage has its own purpose, the three tuning stages can be considered as "independent". Therefore, the tuning process is not too complicated.

According to our observations, for those images that satisfy the definition of salience always achieve persuasive result. In addition, the modifications of parameters generally do not affect the segmentation result for those images. Therefore, after a few

iterations, we got an acceptable set of parameters. Because the parameters are not sensitive for most images, the parameters can be considered as "robust".

B. Experimental Results and Discussions

We have implemented the proposed algorithm on a Pentium 4 PC, 2.66-GHz CPU with 512-MB RAM using graphical user interface under windows XP. The algorithm is very computational efficient. For CIF format images, the average speed is around 0.4 s per image.

Due to the limitation of paper length, we present some test images and segmentation results of both databases to demonstrate the performance of our work. Fig. 8 and Fig. 9 show the

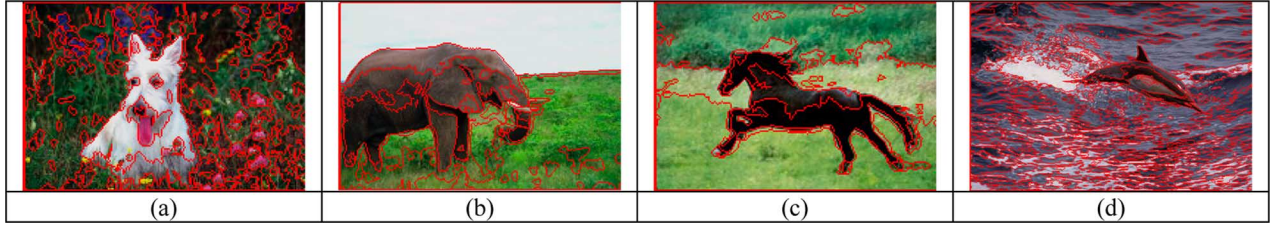


Fig. 11. Segmentation results of the segmenter code of [25].

experimental results of our database and Malik's database, respectively.

It is worth mentioning that our merging rule takes into consideration not only the similarity of region color but also the geometric properties such as boundary length and region size, which make the results more consistent with human perception. As shown in Fig. 9.20(b), the quantized color of the black stripe of the tropical fish is more similar to the color of the coral reef than to that of the yellow and white stripes of the tropical fish. If color is the only consideration, the black stripe should be merged into the coral reef. Nevertheless, we got a complete fish in our result due to considering the boundary length. In addition, as shown in Fig. 9.16, if the color is the only consideration, the zebra will be broken apart by the black and white stripes. The criteria of region size and boundary length force them to group into a region.

In our method, the further checking scheme is a critical mechanism. It increases the reasonability and improves the segmentation result significantly. Initially, according to the importance index, we merge the less important regions one by one. However, when the important index of surviving regions exceeds the threshold T_m , they may still be similar and should further merge. With the further checking mechanism, they have the chance to merge if they are similar, and then achieve results that are more reasonable. Fig. 7(d) shows the intermediate state just after the first stage of merging. There are seven surviving regions. From Table I we notice that the color distances of some regions (with sky-blue) are less than the threshold T_S , and therefore should be further merged. After the further checking, we have the result as shown in Fig. 7(e) and (f).

In the stage of dominant color extraction and quantization, apparently, more color candidates make more precise mapping from original color to quantized color. Thus, if we use a narrower bandwidth to produce more candidates of colors, we have a better result. Nevertheless, the larger the number of candidates results in longer merging time. Therefore, in our work, we use a narrow bandwidth for Y channel and a relatively wider bandwidth for U, V channels to reduce the number of candidates. Because we weight the chromatic channels much more than the intensity channel in the merging stage, it will be harder to merge regions with different chromatic levels. On the contrary, if the chromatic levels are fewer, the similar colors will map to the same level on the quantization stage. Therefore, as our expectation, regions are formed implicitly based on chrominance. Consequently, we save not only the number of candidates but also the processing time with even better result.

Based on above discussion, we summarize two key points about our work as follows:

- 1) In the quantization stage, use more levels on luminance and fewer levels on chrominance making a better mapping from source image to quantized image while preserving the differentiability.
- 2) In the merging stage, weight more in chrominance and less in luminance reducing the sensitivity of luminance.

Besides, too much detail will destroy the salience, therefore, we encourage the region merging as complete as possible. In most case, over-merging is acceptable.

To demonstrate the main difference between the proposed method and the conventional techniques, we downloaded the segmenter code of [25], which applies the mean shift algorithm for segmentation, from their web site [40] and executed with the "Undersegmentation" option. Some segmentation results are shown in Fig. 11. Although, the topic of these images seems very simple, i.e., an apparent object in image, the segmentation results still contain too many details. Hence, it is very difficult to focus on image salience even though the salient region is significant. However, our method successfully achieves the salient region segmentation, which is consistent with human perception. Please refer to Fig. 8.1(d), 8.3(d), 8.8(d) and 8.12(d). In addition, because we are interested in salient regions in image, some region boundaries that do not precisely match the contours of objects are acceptable.

C. Limitation

The proposed algorithm is based on the definition of salience and focuses on color images. If an image is with no specific salient region or with fully texture, it may achieve unsatisfactory segmentation results. Besides, if the chrominance is poorly distributed, i.e., centered in a very narrow range, we may also fail, because there is no differentiability in chrominance.

Fig. 10 shows some typical failure cases. In Figs. 10.1 to Fig. 10.3, the colors of the main objects are almost the same as their background. The creatures perfectly blend themselves into the background. It is very difficult to segment them from the background. In addition, there is no specific salient region in Fig. 10.4. It is hard to determine a good segmentation even for human. In our work, the parameters' setting is suitable for most images. Inevitably, in some cases, the setting may yields unsatisfactory results such as Fig. 10.5, which is an over-merging case. Although it can be easily fixed by using lower T_s , a robust parameters tuning procedure should be further studied.

V. CONCLUSION

We have presented a new salient region segmentation approach for color images based on dominant color extraction and region merging. A nonparametric density estimation was first

employed to extract dominant colors and a fast color mapping algorithm was developed to quantize images in an efficient way. Computation rules of "importance index" and "merging likelihood" of regions were then developed to merge the initial regions generated in the quantization step. Finally, an adaptive threshold is used to further merge those important regions. The proposed approach efficiently extracts salient regions in color images. Experiments show that the segmentation results satisfied our definition of salience, and the proposed method effectively addressed the over-segmentation problem in traditional segmentation algorithms.

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